

Generation Interconnection System Impact Study Report

For Queue Project Q21-1880

Gwf 4 138 kV
March 06, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1880. The Interconnection Customer proposes a 124.2 MW solar photovoltaic facility interconnecting at the Gwf 4 138 kV substation in the West Virginia area. The request was received on September 12, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 570 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,727,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The GWF 4 / NORTH HAVEN 138kV line is overloaded to 120.5% of its 196 MVA emergency rating for the loss of the PENCE / VALEMOUNT SUBSTATION 345kV. The project contributes 55.1% DFAX (68.4 MW) to this overload.

For the loss of VALEMOUNT / PENCE 345kV Line, the N HAVEN / DORRIS 138kV loads to 100.1% of its 220 MVA rating. Project DFAX: 55.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GWF 4 / NORTH HAVEN 138kV line	Rebuild line with higher-capacity conductor	\$4,432,000	66% (\$2,936,000)
N HAVEN / DORRIS 138kV	Reconductor line	\$2,703,000	66% (\$1,791,000)
	Total allocated to project		\$4,727,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.